TIG2: Exact Cubic Branch Structure of Structural Horizon Formation

Kai Stefan Dietrich
Independent Researcher
Topological Integrity Gravity Framework

May 11, 2026

Abstract

Topological Integrity Gravity (TIG) formulates horizon formation as a structural consistency problem rather than merely as a coordinate-dependent feature of a specific metric representation. In this work, we isolate and analyze the exact cubic branch structure associated with the minimal TIG horizon normal form. The central equation is

$$x^3 - x^2 + \beta^3 = 0,$$

where x denotes the dimensionless horizon radius and β is a dimensionless structural deformation parameter. We show that the branch structure contains a critical degeneracy at

$$x_c = \frac{2}{3}, \qquad \beta_c = \left(\frac{4}{27}\right)^{1/3},$$

where two real branches coalesce in a saddle-node transition. The Schwarzschild limit is recovered for $\beta \to 0$, while the critical point defines a structurally distinguished boundary for admissible horizon formation. The result provides a compact and mathematically exact formulation of the TIG2 cubic branch sector and prepares the ground for further analyses of photon spheres, echo delays, and near-critical observables.

1 Introduction

Classical general relativity describes gravitational dynamics through the geometry of spacetime and the Einstein field equations [1–3]. In black-hole physics, horizons are usually treated as geometric or causal structures associated with exact or approximate spacetime solutions [2, 4].

In the Schwarzschild limit, the horizon radius is located at

$$r_H = 2M.$$

Using the dimensionless normalization

$$x := \frac{r_H}{2M},$$

the Schwarzschild branch corresponds to

$$x = 1.$$

Topological Integrity Gravity (TIG) modifies the conceptual starting point. Instead of treating horizon formation only as a derived geometric surface, TIG interprets admissible horizon formation as a structural consistency condition. The present paper builds on the baseline TIG formulation [5] and isolates the exact cubic branch structure of the second TIG sector.

The purpose of this paper is deliberately narrow. We do not propose a full replacement of the Einstein field equations. Rather, we isolate the minimal algebraic normal form governing the TIG2 structural horizon branch and analyze its exact critical geometry. The structural horizon equation considered here is

$$x^3 - x^2 + \beta^3 = 0. \tag{1}$$

Here x denotes the dimensionless horizon radius, while β denotes a dimensionless structural deformation parameter. The parameter β is not interpreted as an additional spatial dimension. It acts as a structural control parameter measuring deformation away from the Schwarzschild branch.

Equation (1) has three central properties. First, the Schwarzschild branch is recovered exactly for $\beta = 0$. Second, the cubic contains a finite critical branch point. Third, its discriminant yields an exact criterion for branch coalescence.

2 Structural Horizon Equation

The TIG2 structural horizon equation is

$$x^3 - x^2 + \beta^3 = 0. \tag{2}$$

For $\beta = 0$, the equation reduces to

$$x^3 - x^2 = x^2(x-1) = 0.$$

Thus the nontrivial Schwarzschild-connected branch is

$$x = 1.$$

The double root at $x = 0$ is interpreted as a degenerate base structure rather than as a macroscopic horizon. The deformation term $+\beta^3$ minimally unfolds the degenerate polynomial

$$P_0(x) = x^2(x-1).$$

This form is useful because it preserves the Schwarzschild limit while introducing a single structural control parameter.

Related questions about horizon regularity, singularity avoidance, and modified high-curvature behavior have appeared in several different contexts, including singularity theorems, regular black-hole models, and higher-curvature gravity [6–10].

3 Critical Point

The critical point occurs when the cubic possesses a repeated root. This requires

$$P(x;\beta) = 0$$

and

$$\frac{\partial P}{\partial x} = 0.$$

From

$$P(x;\beta) = x^3 - x^2 + \beta^3,$$

we obtain

$$\frac{\partial P}{\partial x} = 3x^2 - 2x.$$

Therefore

$$3x^2 - 2x = 0,$$

or

$$x(3x-2) = 0.$$

The nontrivial critical branch occurs at

$$x_c = \frac{2}{3}.$$

Substitution into the cubic yields

$$\left(\frac{2}{3}\right)^3 - \left(\frac{2}{3}\right)^2 + \beta_c^3 = 0.$$

Hence

$$\frac{8}{27} - \frac{12}{27} + \beta_c^3 = 0,$$

which gives

$$\beta_c^3 = \frac{4}{27}.$$

Therefore,

$$x_c = \frac{2}{3}, \qquad \beta_c = \left(\frac{4}{27}\right)^{1/3}. \tag{3}$$

This point corresponds to the exact saddle-node coalescence of two real cubic branches.

4 Discriminant Structure

For a cubic polynomial

$$ax^3 + bx^2 + cx + d = 0,$$

the discriminant is

$$\Delta = b^2c^2 - 4ac^3 - 4b^3d - 27a^2d^2 + 18abcd.$$

For the TIG2 cubic

$$x^3 - x^2 + \beta^3 = 0,$$

the coefficients are

$$a = 1, \qquad b = -1, \qquad c = 0, \qquad d = \beta^3.$$

The discriminant therefore becomes

$$\Delta = 4\beta^3 - 27\beta^6.$$

Equivalently,

$$\Delta = \beta^3(4 - 27\beta^3). \tag{4}$$

The branch classification follows immediately:

- for $0 < \beta^3 < 4/27$, one has $\Delta > 0$, corresponding to three distinct real roots;
- for $\beta^3 = 4/27$, one has $\Delta = 0$, corresponding to branch coalescence;
- for $\beta^3 > 4/27$, one has $\Delta < 0$, leaving only one real branch.

The Schwarzschild-connected positive branch terminates precisely at the critical point.

5 Branch Geometry

The cubic equation may also be written as

$$x^2(1 - x) = \beta^3. \tag{5}$$

For positive β, the right-hand side is positive, implying

$$x^2(1 - x) > 0.$$

For positive x, this requires

$$0 < x < 1.$$

Thus the Schwarzschild branch begins at $x = 1$ for $\beta = 0$ and moves inward as the structural deformation parameter increases.

Define

$$f(x) = x^2(1 - x).$$

Its derivative is

$$f'(x) = 2x - 3x^2 = x(2 - 3x).$$

The positive maximum occurs at

$$x = \frac{2}{3},$$

where

$$f\left(\frac{2}{3}\right) = \frac{4}{27}.$$

This reproduces the exact critical value

$$\beta_c^3 = \frac{4}{27}.$$

The critical point therefore has a direct geometric interpretation: it corresponds to the maximum structural capacity of the admissible branch geometry.

6 Exact Cubic Branch Structure

Figure 1 illustrates the exact cubic branch structure associated with the TIG2 horizon equation.

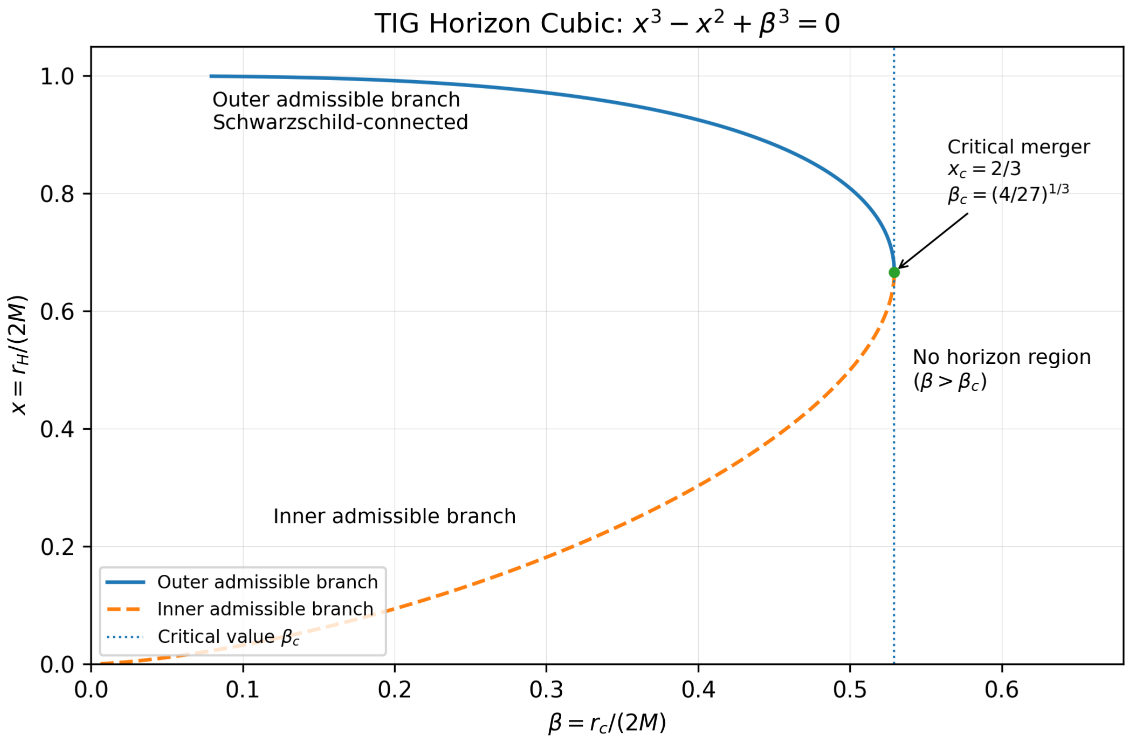

Figure 1: Exact cubic branch structure of the TIG2 structural horizon equation $x^3 - x^2 + \beta^3 = 0$. The Schwarzschild branch begins at $x = 1$ for $\beta = 0$ and terminates at the critical saddle-node point

$$x_c = \frac{2}{3}, \qquad \beta_c = \left(\frac{4}{27}\right)^{1/3}.$$

7 Near-Critical Expansion

To study the local structure near the critical point, define

$$x = x_c + \epsilon, \qquad \beta^3 = \beta_c^3 - \delta,$$

with

$$x_c = \frac{2}{3}, \qquad \beta_c^3 = \frac{4}{27}.$$

Expanding

$$P(x;\beta) = x^3 - x^2 + \beta^3$$

around the critical point yields

$$P = \epsilon^2 + \epsilon^3 - \delta.$$

To leading order,

$$\epsilon^2 \simeq \delta.$$

Therefore,

$$\epsilon \simeq \pm\sqrt{\delta}. \tag{6}$$

The TIG2 branch splitting thus obeys universal saddle-node square-root scaling.

8 Physical Interpretation

The TIG2 cubic sector does not claim to represent a complete covariant gravitational theory. Its role is structural. The physical interpretation is intentionally conservative:

1. the Schwarzschild limit is exactly recovered;
2. the parameter β introduces a controlled structural deformation;
3. a finite critical point emerges naturally;
4. the branch termination is mathematically exact;
5. near-critical behavior follows universal saddle-node scaling.

The cubic sector therefore defines a mathematically rigorous structural core from which more detailed metric and observational sectors may later be derived. This interpretation is compatible with the broader lesson from classical relativity: exact solutions, causal horizons, and singularity formation must be treated with care, especially in regimes where extrapolation beyond established curvature scales becomes physically delicate [2, 4, 6].

9 Why a Cubic?

The cubic structure is not chosen arbitrarily. It represents the minimal nonlinear algebraic sector capable of generating a finite branch coalescence with a nontrivial discriminant structure.

A quadratic deformation does not produce a finite saddle-node branch transition with a structurally distinguished critical point. By contrast, the cubic equation naturally generates:

- exact branch splitting,

- finite discriminant collapse,
- critical branch coalescence,
- universal near-critical scaling.

The TIG2 cubic horizon equation therefore represents the minimal algebraic normal form capable of supporting a structurally nontrivial horizon transition sector.

This places the TIG2 structure in close conceptual relation to catastrophe theory and nonlinear bifurcation systems, where cubic normal forms frequently govern universal transition behavior.

10 Universality of the Critical Scaling

The near-critical expansion produces the local scaling relation

$$\epsilon \sim \sqrt{\delta}.$$

This square-root behavior is not tied to a particular metric ansatz or microscopic realization. Instead, it follows universally from the local catastrophe structure of the cubic horizon sector itself.

The resulting scaling law therefore reflects a structural property of the branch geometry rather than a model-dependent dynamical assumption.

In this sense, the critical TIG2 branch structure behaves analogously to universal saddle-node transitions encountered throughout nonlinear systems and bifurcation theory.

11 Observable Outlook

The present paper intentionally isolates only the structural cubic sector. Nevertheless, the resulting branch geometry suggests several potentially observable near-critical consequences.

These include:

- modified photon-sphere radii,
- altered effective near-horizon potentials,
- near-critical echo-delay amplification,
- branch-sensitive shadow deviations,
- critical deformation signatures in high-curvature regimes.

The purpose of the present work is not to derive these observables in full detail, but rather to establish the mathematically exact structural core from which such analyses may consistently proceed.

12 Relation to General Relativity

General relativity is recovered in the limit

$$\beta \to 0.$$

In this limit,

$$x \to 1,$$

which corresponds to

$$r_H \to 2M.$$

The TIG2 cubic sector should therefore not be interpreted as a contradiction of general relativity. Instead, it provides an additional structural criterion that becomes relevant near critical branch configurations.

This places TIG2 closer in spirit to structural or effective high-curvature analyses than to a direct replacement of classical general relativity. The comparison class includes higher-curvature modifications such as the R^2 model of Starobinsky [8], regular black-hole constructions [7, 9, 10], and standard geometric formulations of relativity [2, 3].

13 Limitations

The present work is intentionally limited in scope. Specifically, the paper does not:

- derive a full covariant action principle;
- propose a complete modified Einstein equation;
- derive observational constraints;
- interpret β as a spatial coordinate;
- claim that the cubic sector alone determines all black-hole observables.

The purpose of TIG2 is instead to establish a mathematically exact structural core suitable for future extension.

14 Conclusion

We have analyzed the exact cubic branch structure associated with the TIG2 structural horizon equation

$$x^3 - x^2 + \beta^3 = 0.$$

The cubic reproduces the Schwarzschild limit exactly and contains a finite saddle-node critical point at

$$x_c = \frac{2}{3}, \qquad \beta_c = \left(\frac{4}{27}\right)^{1/3}.$$

Its discriminant is

$$\Delta = \beta^3(4 - 27\beta^3),$$

which defines the precise branch transition structure.

Near criticality, the branch splitting obeys square-root scaling, characteristic of universal saddle-node behavior. The result establishes a compact and mathematically transparent structural core for TIG2 and provides a stable basis for future analyses of effective geometries, photon-sphere deviations, echo structures, and observational near-critical signatures.

Acknowledgments

The author acknowledges the broader Topological Integrity Gravity research program from which the present structural analysis emerged.

References

[1] Albert Einstein. Die feldgleichungen der gravitation. *Sitzungsberichte der Preussischen Akademie der Wissenschaften zu Berlin*, pages 844–847, 1915.

[2] Robert M. Wald. *General Relativity*. University of Chicago Press, 1984.

[3] Sean M. Carroll. *Spacetime and Geometry*. Addison Wesley, 2004.

[4] Stephen Hawking and George Ellis. *The Large Scale Structure of Space-Time*. Cambridge University Press, 1973.

[5] Kai Stefan Dietrich. Topological integrity gravity. *arXiv preprint*, 2026. TIG1 baseline paper.

[6] Roger Penrose. Gravitational collapse and space-time singularities. *Physical Review Letters*, 14 (3):57–59, 1965.

[7] James M. Bardeen. Non-singular general relativistic gravitational collapse. *Proceedings of GR5*, 1968.

[8] A. A. Starobinsky. A new type of isotropic cosmological models without singularity. *Physics Letters B*, 91(1):99–102, 1980.

[9] Irina Dymnikova. Vacuum nonsingular black hole. *General Relativity and Gravitation*, 24(3): 235–242, 1992.

[10] Sean A. Hayward. Formation and evaporation of non-singular black holes. *Physical Review Letters*, 96:031103, 2006.

[11] Eloy Ayón-Beato and Alberto García. Regular black hole in general relativity coupled to nonlinear electrodynamics. *Physical Review Letters*, 80(23):5056–5059, 1998.